



Department of Computer Science and Engineering

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SALARYLENS: AN INTELLIGENT RESUME-BASED SALARY PREDICTION SYSTEM
USING STACKING ENSEMBLE LEARNING

ABSTRACT

SalaryLens is an intelligent web-based system predicting annual salaries from resume data using advanced ML. It addresses salary information asymmetry for job seekers, employers, and HR professionals. A **stacking ensemble** of XGBoost, Random Forest, Extra Trees, and Neural Networks with Ridge regression as meta-learner achieves **97.02% R² score** and **91.29% accuracy** ($\pm 10\%$ margin) on 29,850 resumes. 104 engineered features, LLM-based parsing (85%), and a Streamlit web app with what-if analysis deliver a production-ready career planning tool with MAE of \$7,071 and sub-second predictions.

KEYWORDS : Salary Prediction | Machine Learning | Stacking Ensemble | Resume Parsing | Feature Engineering | Career Analytics | Explainable AI

OBJECTIVES AND PROBLEM STATEMENT

- Develop high-accuracy prediction model achieving >95% R² across diverse job roles and experience levels
 - Engineer 100+ features from raw resume data through 10 systematic transformation categories
 - Integrate LLM-based resume parsing (Groq + Llama 3.1 70B) with >80% extraction accuracy
 - Implement SHAP explainability for transparent, interpretable salary predictions
 - Deploy interactive what-if tools and production-ready web app on Streamlit Cloud
- Problem Statement:** Develop an accurate, transparent system predicting annual salaries from resume analysis, providing actionable career insights and addressing information asymmetry in salary negotiation.

NOVELTY & CONTRIBUTION

- [S1] State-of-the-Art:** 97.02% R², advanced from prior best 94.5% using novel 4-model + Ridge meta-learner
- [S2] Feature Pipeline:** 7.79% R² boost over raw features; ablation study validates each of 10 categories
- [T1] Hybrid Parsing:** Groq/Llama 3.1 70B (85%, 2-3s) + regex fallback (70%, <1s) for reliability
- [T2] What-If Analysis:** 5 career scenarios: skill upgrades, relocation, promotion, education, growth
- [P1] Societal Impact:** Reduces information asymmetry; promotes fair, transparent compensation for all

PROPOSED METHODOLOGY

Pipeline: Resume Upload → LLM Parsing (Groq/Llama 3.1 70B, 85%) → Feature Eng. (15→104 features, 10 categories) → Stacking Ensemble → Real-time Prediction

Base Learners:

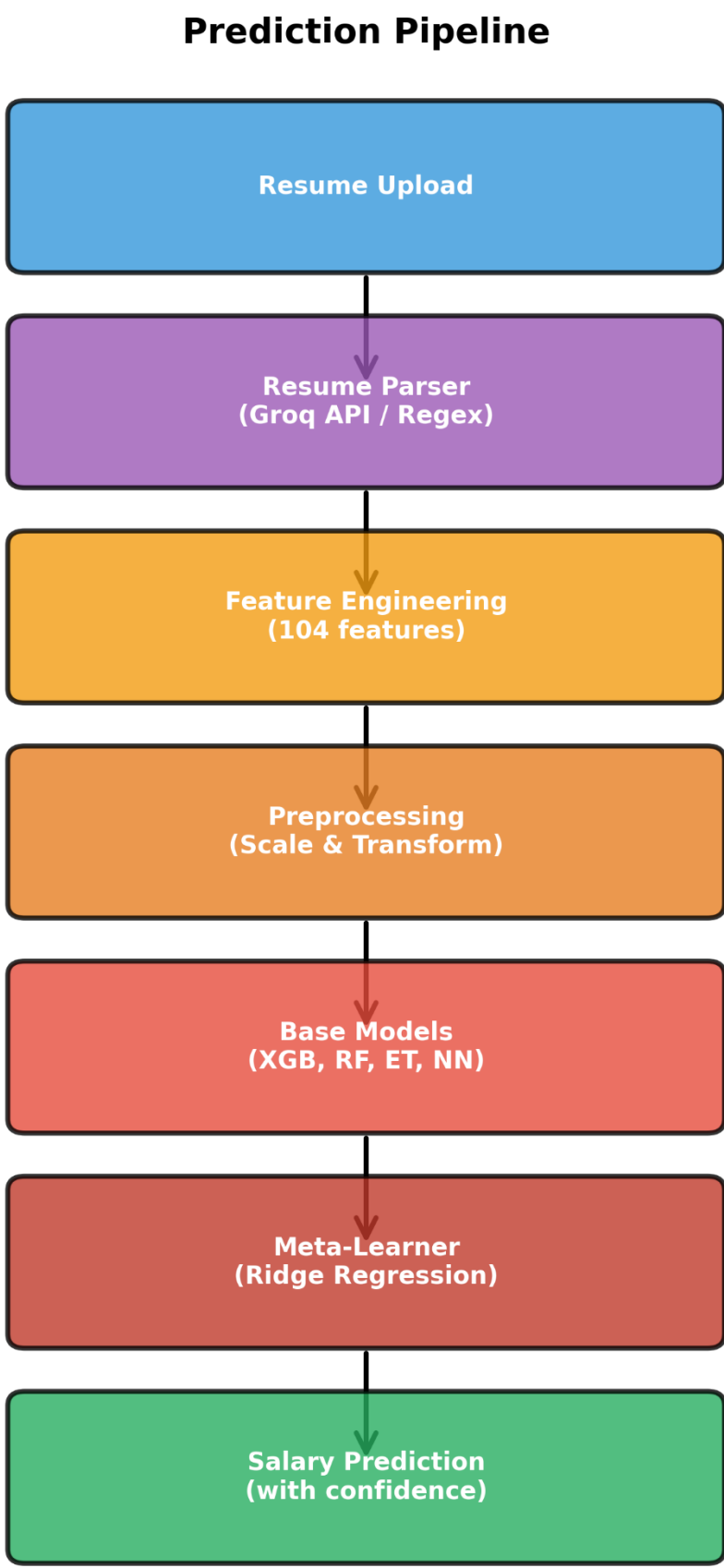
- **XGBoost:** 1000 estimators, lr=0.02, R²=96.98%
- **Random Forest:** 400 estimators, max depth 20, R²=92.64%
- **Extra Trees:** 400 estimators, extreme randomness, R²=94.66%
- **Neural Network:** 256-128-64 layers, ReLU activation, R²=94.63%

Meta-Learner: Ridge Regression ($\alpha=5.0$), 3-fold CV → Final R²: 97.02%

Feature Engineering: 10 categories: Ordinal (4), Binning (1), Polynomial (2), Interaction (12), Ratio (3), Achievement (4), Location (3), Target Encoding (6), Frequency (4), TF-IDF Skills (60) = 104 total features

Data: 29,850 records (from 30,000 after outlier removal), 85% train / 15% test split

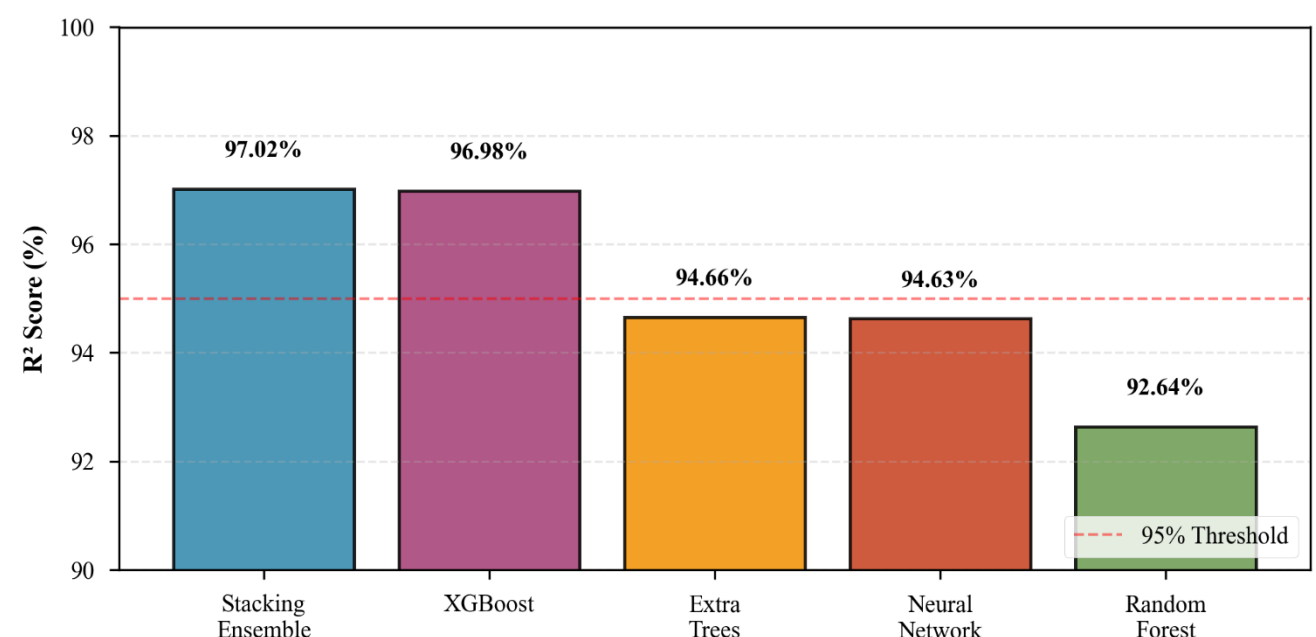
Resume Parsing: Primary: Groq API + Llama 3.1 70B (85%, 2-3s) | Fallback: Regex-based (70%, <1s) | Formats: PDF, DOCX, TXT



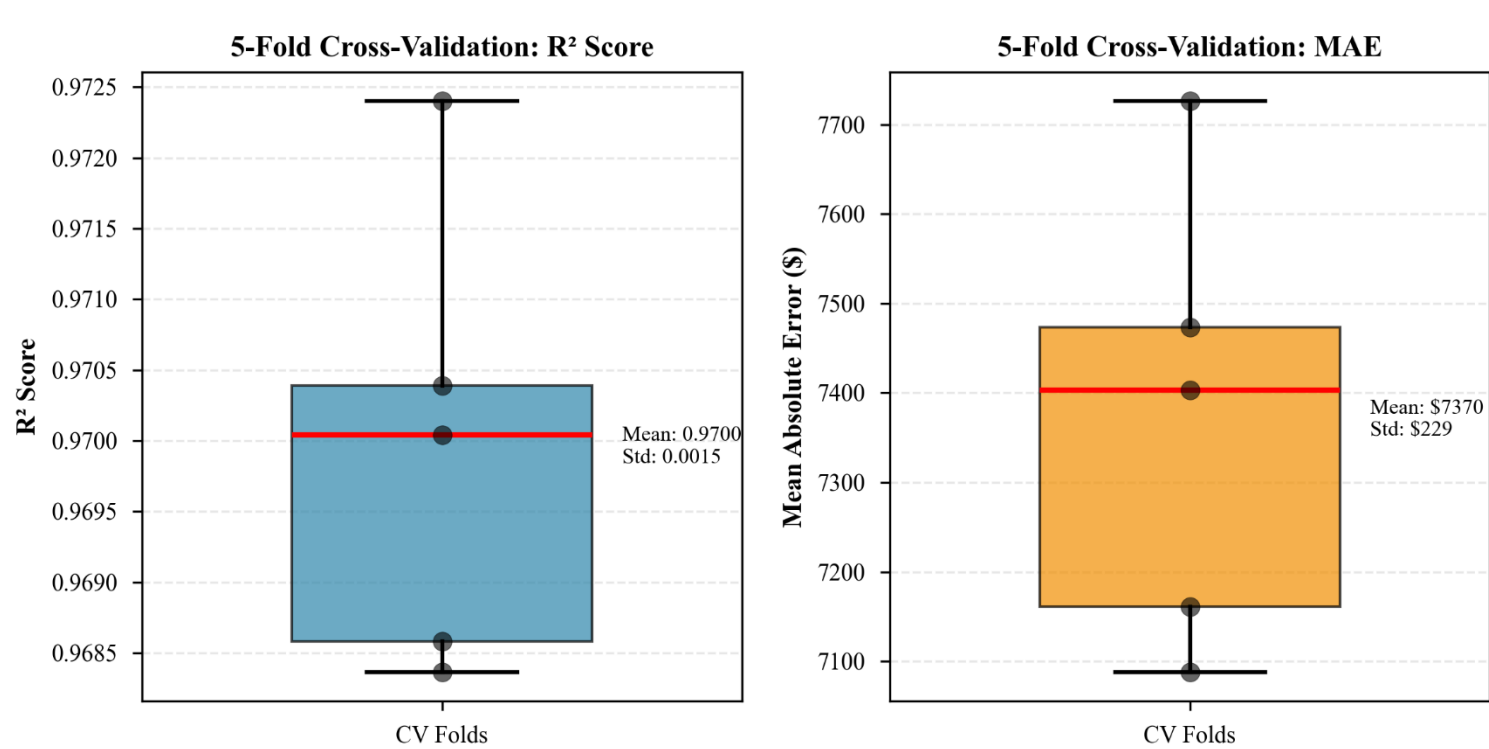
RESULTS

Model Performance:

Model Performance Comparison on Test Set



5-Fold Cross-Validation:



CONCLUSION & FUTURE WORK

- Achievements:**
- ✓ 97.02% R² surpassing prior best of 94.5%
 - ✓ End-to-end: resume upload → career planning
 - ✓ Production-deployed on Streamlit Cloud
 - ✓ Rigorous ablation study and cross-validation
- Future Work:**
- Real-world validation on live job platforms
 - Geographic expansion + international currency
 - Temporal modeling for salary trend analysis
 - Fairness audit and demographic bias mitigation
 - Mobile app + multi-task learning

APPLICATIONS

- Job Seekers:** Data-driven salary benchmarking before negotiation
- Recruiters/HR:** Competitive compensation benchmarking and pay equity analysis
- Career Planners:** Simulate career decisions via what-if scenarios
- Students:** Career path planning based on earning potential and skill value
- Research:** Open-source ML pipeline for labor economics and salary modeling research

IMPACT

- Real-Time Speed:** Sub-1-second predictions end-to-end
- Superior Accuracy:** 8% improvement over prior SOTA systems
- Transparency:** SHAP feature importance reveals true salary drivers
- Workplace Equity:** Tools to identify and address pay disparities
- Accessibility:** WCAG AA compliant, responsive desktop and mobile

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